F1: symmetric membership and generic torus cosets on P-256

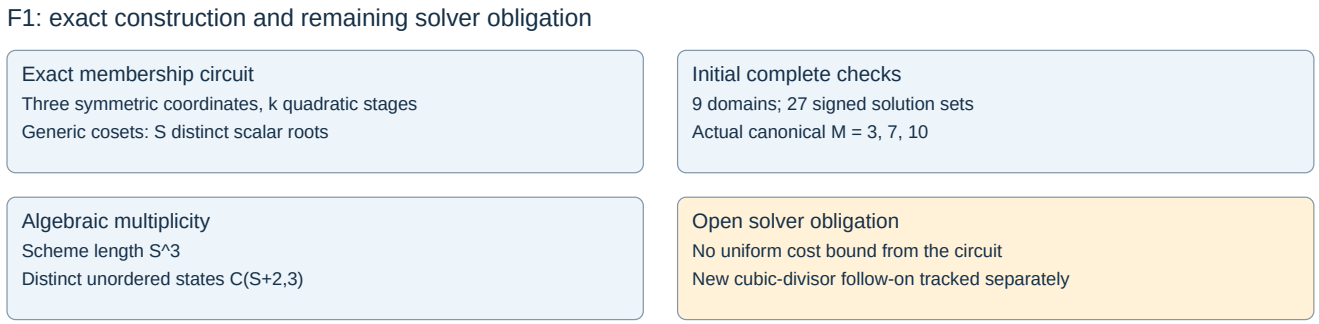

Figure 1: Editable study diagram; exact scope and evidence paths appear in the report.

2026-10-25 · TASK-20261025-cfb933 · implementation validation and exact mathematical screening. This is neither an official performance trial nor a hypothesis transition. No EXP identifier, speed advantage, or complete ECDLP claim is made.

The delivered symmetric quadratic membership circuit works over the published P-256 field. Two actual algebraic solution formulations were exercised. Complete finite-domain decomposition sets pass a separate replay implementation. Global implicit solving remains open: backward recursion enumerates the full triple space, while expanded forward elimination already needs explicit handling of nonreduced symmetric fibres.

Fixed inputs and environment

The input commit supplied by the Coordinator is 2c8c71cfff6f7f63297fbe4de4608cd94d857184. Exact committed constants and the current 35-term S4 polynomial are loaded from baseline/parameters.py and baseline/s4-symmetric.json; their SHA-256 digests are in checks-02/checks.json (checks-02/checks.json). P-256 means

```
p = 2^256 − 2^224 + 2^192 + 2^96 − 1
a = −3
b = 0x5ac635d8aa3a93e7b3ebbd55769886bc651d06b0cc53b0f63bce3c3e27d2604b
r = 0xffffffff00000000ffffffffffffffffbce6faada7179e84f3b9cac2fc632551
h = 1.
```

The implementation uses /workspace/dickson-solver/.venv/bin/python: CPython 3.12.14, SymPy 1.14.0, python-flint 0.8.0. The repository-prescribed /usr/local/bin/python3 is absent; that failed probe is retained. No installation, Docker use, Git action, PR edit, or shared-file edit was performed. The shared working tree is managed by the parent session; this track adds its scoped source and artifacts and does not assert a clean tree. Runtime model binding is inherited and unprobed. Hardware timing and peak memory were not measured: resource watchdog limits and algebraic operation/state counts are reported, without a timing study.

Derived construction

For elementary symmetric coordinates (u,v,w) of three input coordinates, one Dickson step $x \mapsto x^2-2$ induces

$$\Phi(u,v,w) = (u^2-2v-6,\ v^2-2uw-4u^2+8v+12,\ w^2-2v^2+4uw+4u^2-8v-8).$$

The terminal equations are $\Phi^k(u,v,w)=(3c,3c^2,c^3)$, joined to S4. Generic terminals are $c=\mathrm{Tr}(z_0^{(2^k)})$, $c\neq\pm2$, for z_0 in the norm-one torus over $\mathbb{F}_{p^2}$. Every exact-order power-of-two torus generator is certified by exponentiation. Generic domains have 2^k distinct coordinates; ramified $c=2$ domains have $2^{(k-1)}+1$. Actual factor-base sizes M count only canonical x-coordinates lifting to P-256 points. Affine controls use $x=3t+\beta$ and the corresponding symmetric-coordinate transformation.

derivations.md (derivations.md) gives the proofs, formulas, scope and accounting. diagram.svg (diagram.svg) is editable vector source; the Coordinator owns the aggregate PDF rendering. This track changes no shared canonical graph.

Checked observations

The complete finite-domain workload was fixed to ramified k=3,4,5, with one generic k−1 coset and one affine coordinate control per size. A deterministic bounded coset search and bounded β search matched the actual lifted M; every search candidate is retained in the receipt. These chosen matches are correctness fixtures, not a distributional sample.

Actual M	Ramified S / domain N / states	Generic S / domain N / states	Affine generic states
3	8 / 5 / 35	4 / 4 / 20	20
7	16 / 9 / 165	8 / 8 / 120	120
10	32 / 17 / 969	16 / 16 / 816	816

Here states means exactly C(N+2,3) symmetric coordinate triples before curve lifting. All nine bundles supply their full domain, signed factor base, targets, all accepted signed decomposition index triples, branch counts and solution hashes. Each has three finite public target checks:

- A deliberately constructed positive, with one complete signed decomposition.
- A base-member target, with exactly M decompositions from inverse-pair

cancellation; these certify exceptional-case handling and are not counted as independent useful relations.

- The published P-256 generator G as a fixed unrelated point, with a certified

empty decomposition set in that finite domain.

All 27 sets match both a signed point pair join and an independently implemented exhaustive signed-triple replay. The replay also verifies explicit full polynomial factor identities for each domain's squarefree radical, and exact lifted-base membership. This is separate code in the same session, not independent review. No generation scalar is passed to the solvers. Positive fixtures certify correctness; the empty G checks say nothing about uniform-target success rates.

Additional exact checks comprise three symbolic circuit identities, 40 numeric circuit identities, 40 S4/resultant identities, the published generator order, every retained base point's order, and seven torus/circuit witness depths k=6,7,8,16,32,64,96. Complete generic domains at the first three depths have sizes 64,128,256 and actual M=29,58,119. The four higher depths have witnesses only, without domain or relation-solution completeness claims.

Forward algebraic formulation

The forward adapter expands three membership polynomials, adjoins S4, computes a grevlex Groebner basis, then recursively obtains rational roots via lex elimination and FLINT univariate roots. Source and input equations are retained, including all finite symmetric solutions on completed joint cases.

Fixed check	Quotient length	Distinct tested zero states	Outcome
Membership k=1	8	4	Complete membership reference
Membership k=2	64	20	Complete membership reference
Membership + S4 k=1	3	2	All rational roots match inverse reference
Membership + S4 k=2	2	1	All rational roots match inverse reference
Membership + S4 k=3	unavailable	unavailable	30-second process watchdog during basis construction

These forward fixtures are subfibres of the fixed generic k=3 case; they are identified by their exact c and target point in their files. They are distinct from the matched-M fixture selection. All five attempts share a declared 2 GiB address-space limit and are run sequentially. The stopped case is a resource outcome for this SymPy adapter, not a certified empty set or a mathematical obstruction. Groebner ideal-membership witnesses are not exported; completed joint answers are checked against a separate exhaustive finite-domain reference.

A notable exact issue survives switching to an unramified scalar coset: the symmetric membership scheme is nonreduced at the triple-equal terminal. Its length is S^3 but its distinct points number C(S+2,3), with local multiplicities 1,3,6 according to repetition type. The measured finite quotient lengths agree with the derived formula. A solver using a squarefree-characteristic-polynomial certificate on the unradicalized algebra cannot be applied without adaptation.

Growth and operation counts

A shared Φ stage uses 3 squarings, 1 general multiplication, 8 written small-scalar scalings and 11 additions/subtractions. An auxiliary-variable representation has 3k quadratic equations in 3(k+1) variables plus terminal equations and S4. No bound on its elimination or solving degree has been established.

Expansion produces degree 2^k. Exact combined nonzero coefficient slots for the three outputs at k=1…5 are 14,42,180,1008,6664. The corresponding per-output counts and a dense total-degree cap are in representation-01.json (representation-01.json). The backward solver has at most eight raw inverse branches per state and stores C(N+2,3) complete leaf states; actual square-root requests and raw branches at every level appear in the case files. It avoids dense pair polynomials but enumerates triples. These are algebraic representation counts, not timing comparisons or lower bounds on every implicit algorithm.

Commands, failures and limits

Run the following from /workspace/crypto-autoresearcher; use a fresh output folder for a replay, retaining existing records unchanged:

```
PY=/workspace/dickson-solver/.venv/bin/python
TRACK=research/p256_factor_base_parallel_20261025/tracks/f1
$PY "$TRACK/f1.py" --out "$TRACK/replay-new"
$PY "$TRACK/verify.py" "$TRACK/checks-02"
$PY "$TRACK/representation.py"
```

execution.json (execution.json) records exact original commands, environment, source hashes and failure paths. check_forward.py contains the five exact adapter commands and fixed watchdog; its output directory must be new. forward-01/receipt.json (forward-01/receipt.json) records every terminal outcome, with native stdout/stderr per case. An initial implementation attempt incorrectly chose the order-four torus element z(1)=i for high-depth generic checks, causing c=2 and an assertion failure. Its source, console and nine already-written finite bundles remain in attempts/ and checks-01/; corrected checks-02 is separate.

The complete-solution certificates cover only the declared finite domains on actual P-256. No hidden base logarithms, relation rank, online success distribution, linear algebra, final target attachment, or end-to-end costs were established. No asymptotic bound for all implicit solvers follows from the inverse enumeration or the bounded forward watchdog.

The precise remaining obligation is recorded in problem-implicit-global-solver.md (problem-implicit-global-solver.md): implement and certify radical/local-algebra extraction on the unexpanded chain, avoiding both complete domain-triple enumeration and per-base dense pair polynomials, then give a justified total-cost model. The fixed k=1,2,3 public fixtures are

immediate controls for that follow-on work. Root indexing and archival are the Coordinator's scope. Neither publication nor completion of this implementation pass changes an official research status.